

KUMAR GAURAV

+91 8210765395 kr.gaurav931@gmail.com Bhubaneswar, Odisha
linkedin.com/in/grv931/ github.com/grv931 leetcode.com/grv931 Portfolio

Education

International Institute of Information Technology Bhubaneswar **2023 – 2027**
Bachelor of Technology in Electrical and Electronics Engineering *CGPA: 8.2/10*

Indian Institute of Technology Madras **2023 – 2027**
Bachelor of Science in Data Science and Applications (Remote) *CGPA: 7.2/10*

- Coursework: Programming, Data Structures and Algorithms, Database Management Systems, System Commands, Machine Learning, Deep Learning and Generative AI

Projects

Household Services Application - SkillFleet | *Flask, VueJS, SQLite, Redis, Celery* 

- Built a full-stack household services marketplace connecting customers with verified service professionals through a unified booking system.
- Designed role-based workflows for Admin, Customer, and Professional users, supporting service requests, approvals, reviews, and status tracking.
- Leveraged Redis caching and Celery task queues to automate notifications, report generation, and large-scale data export operations.

Agri-Tech Image Super-Resolution Model | *Deep Learning, cGAN, PyTorch* 

- Built a cGAN-based image super-resolution pipeline to upscale degraded crop leaf images by 4x.
- Leveraged adversarial and perceptual learning to restore disease-specific visual patterns and improve reconstruction fidelity.

Thoracic Pathology Detection Model | *Deep Learning, Computer Vision, PyTorch, Keras* 

- Built a transfer learning pipeline by fine-tuning DenseNet121 on 51K+ chest X-ray images to classify 20 mutually exclusive thoracic pathologies using Keras.
- Engineered a custom Asymmetric Loss ($\gamma^- = 4$, $\gamma^+ = 1$) with class weighting to combat 66% class imbalance and align training with a cost metric penalising False Negatives 5× over False Positives.

Comment Category Prediction | *Machine Learning, NLP, Python* 

- Built an NLP classifier for comment category prediction using textual and user-engagement features.
- Engineered feature extraction pipelines to improve classification performance on a benchmark dataset.

Technologies

Languages: Python, C, C++, SQL

Frameworks & Libraries: PyTorch, Scikit-learn, Flask, VueJS, SQLAlchemy, NumPy, Pandas

Tools & Databases: Redis, SQLite, Git, GitHub, VS Code, MATLAB

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Machine Learning, Deep Learning, Computer Vision

Achievements

- Qualified GATE 2026 in Data Science and Artificial Intelligence (DA) (Feb 2026).
- Co-authored a conference paper titled "Enhancing Urban EV Charging Networks: A Case Study of Bhubaneswar" utilizing QGIS, Gaussian Mixture Models (GMM), and Particle Swarm Optimization (PSO) (June 2026).
- Secured a top 30 position among 100+ competing teams in the selective CraftNCode Hackathon during the D3 Fest at IIIT Bhubaneswar (Nov 2024).

Extracurricular & Volunteering

- Content & Graphics Lead, E-Cell, IIIT Bhubaneswar; drove branding, marketing, sponsorship outreach, and event operations to execute the E-Summit 2025.
- Earned NCC 'A' and 'B' Certificates under 38 Bihar Battalion.